Inequalities in one triangle

1

a 180°

b The longest side

c The shortest side

2 $\overline{BC}$

3 $\angle ABC$

4 $\overline{AC}$

5 $\angle ABC$

6

a $\overline{AC}, \overline{AB}, \overline{BC}$

b $AC < AB < BC$ or $BC > AB > AC$

7

a $\angle BCA, \angle CAB, \angle ABC$

b $m\angle ABC > m\angle CAB > m\angle BCA$ or $m\angle BCA < m\angle CAB < m\angle ABC$

8

a $x = 8$

b 68°

c 54°

d 58°

e $\overline{BC}$

Additional questions

9 The longest side is opposite the largest angle and the shortest side is opposite the smallest angle.

10

a If one side of a triangle is longer than another side, then the angle opposite the longer side is larger than the angle opposite the shorter side.

b If one angle of a triangle is larger than another angle, then the side opposite the larger angle is longer than the side opposite the smaller angle.

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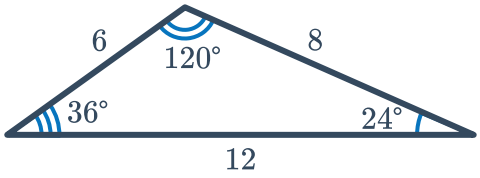
11 **a** $\overline{AC}, \overline{AB}, \overline{BC}$ **b** $\overline{AC}, \overline{AB}, \overline{BC}$

c $\overline{BC}, \overline{AB}, \overline{AC}$ **d** $\overline{AC}, \overline{AB}, \overline{BC}$

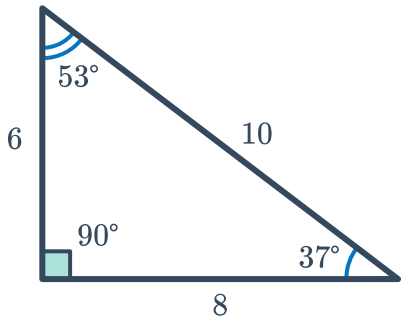
12 **a** $\angle ACB, \angle BAC, \angle ABC$ **b** $\angle ACB, \angle ABC, \angle BAC$

c $\angle ABC, \angle BAC, \angle ACB$ **d** $\angle ACB, \angle ABC, \angle BAC$

13 **a**



b



14 We have the theorem that, "if one side of a triangle is longer than another side, then the angle opposite the longer side is larger than the angle opposite the shorter side". Since we know the hypotenuse is the longest side, this means that the right angle is larger than the other two angles as they are opposite shorter sides. Since these two angles are smaller than the right angle, they are less than 90°, so they are acute.

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